

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – III Examination – Nov 2011

PT202 ELECTROTHERAPY I

Date: 28.11.11, Monday

Time: 10.00 a.m to 10.20 a.m

Maximum Marks: 10

Instructions:

1. There are two parts in the paper. **PART – I** contains 10 Questions and **PART – II** contains 2 Questions.
2. **PART – I** should be answered in MCQ 'S question paper & **PART – II** should be answered in answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 20 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 20 min, can collect main question paper for **PART – II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs)
(Encircle the correct answer)

(1 × 5 = 5)

1. The therapeutic range of Ultra Sound is _____
 - a. 1 – 5 Hz
 - b. 5 – 10 Hz
 - c. 0.5 – 5 MHz
 - d. 1 – 3 MHz
2. UVB and UVC are absorbed in _____
 - a. Cornea
 - b. Lens of Eye
 - c. Both of above
 - d. None of the above
3. Dissipation of heat in SWD is maximized due to _____
 - a. evaporation of sweat
 - b. air circulation
 - c. increased blood flow
 - d. contraction of muscle
4. The depth of penetration of Microwave is _____
 - a. Greater than Shortwaves
 - b. Lesser than Infrared
 - c. Greater than Infrared and lesser than Shortwaves
 - d. None of the above
5. Superluminous diodes in LASER therapy is characterized by _____
 - a. Monochromatic, collimated, coherent
 - b. Nonmonochromatic, collimated, coherent
 - c. Monochromatic, noncollimated, coherent
 - d. Monochromatic, collimated, non-coherent

6. The distance between any two adjacent antinodes or nodes is equal to _____
- λ
 - $\frac{1}{2}\lambda$
 - 2λ
 - None of the above
7. In pulsed ultrasound the thermal effect for _____
- 1:1 and 1:4 produces equal heat
 - 1:4 more heat than 1:1
 - 1:4 less heat than 1:1
 - None of the above
8. $H = I^2RT$ is _____
- Vant Hoff's law
 - Joules's law
 - Ohm's law
 - Kirchoff's law
9. The center of the cable electrode sets up _____
- Magnetic field
 - Electrostatic field
 - Electromagnetic field
 - Both field gets interchanged
10. Metal surfaces or any metal placed on tissue surfaces strongly _____ microwaves.
- Absorb
 - Refract
 - Reflect
 - Diffract

CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY CHARUSAT CAMPUS CHANGA	
Examination Semester _____	
Subject _____ No. _____	
Candidate's Seat No. _____	
Section _____	
Date _____	
	Sr. Supervisor's Sign.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – III Examination – Nov 2011

PT202 ELECTROTHERAPY I

Date: 28.11.11, Monday

Time: 10.20 a.m to 1.00 p.m

Maximum Marks: 60

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 40min.
5. Draw the figure wherever necessary.

PART-II

SECTION –I

Q.1. Long Question (1 out of 2)

(1× 10= 10)

Describe test dose of Ultra Violet Radiation. How will you treat a patient with Acne with Ultra Violet Radiation?

OR

Discuss laws governing Electro Magnetic Radiation and its applications.

Q.2. Write short notes on: (4 out of 5)

(4× 5 = 20)

1. Cluster Probe.
2. Moist heat Vs Dry heat.
3. Paraffin Wax Bath.
4. Production of Micro Wave Diathermy.
5. Whirlpool bath.

SECTION –II

Q.1. Long Question (1 out of 2)

(1× 10= 10)

Discuss production of Ultra sound waves. Discuss Physiological effects and Therapeutic uses of Ultrasound

OR

Describe technique of application of Short Wave Diathermy. How will you treat a female patient with Pelvic inflammatory disease with Short Wave Diathermy?

Q.2. Write short notes on: (4 out of 5)

(4× 5 = 20)

1. Capacitive Energy Transfer.
2. Treatment of Adherent scar using Phonophoresis.
3. PUVA.
4. Treatment of open wound using LASER.
5. PEME.